

IT/OT Convergence and Cybersecurity: Challenges for Data Engineers

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As digital transformation accelerates, the boundary between Information Technology (IT) and Operational Technology (OT) is rapidly dissolving. While this brings significant connectivity benefits, it also introduces serious cybersecurity risks for critical infrastructure. For a Data Engineer working at this intersection, understanding these risks is no longer optional.

IT vs. OT: Different Worlds, Different Priorities

IT systems — servers, laptops, printers — prioritize confidentiality, integrity, and availability, and are patched regularly. OT systems govern physical processes through SCADA, PLCs, and HMIs, and are designed to run continuously for decades in harsh environments (Westermo Network Technologies AB, 2026). Safety and availability take precedence over confidentiality, and downtime is rarely an option — making security updates far more complex than in IT.

The Risks of IT/OT Convergence

Connecting IT and OT networks exposes both to threats neither was designed to handle. The 2021 Colonial Pipeline ransomware attack illustrates this clearly: attackers entered via IT, yet the operator shut down OT pipeline operations as a precaution, triggering major fuel shortages across the eastern United States (Westermo Network Technologies AB, 2026). No OT system was directly breached, yet the real-world consequences were severe.

Holistic Cybersecurity Challenges

Westermo identifies five threat categories: human-influenced events, criminal activity, maintenance gaps, environmental factors, and supply chain risks (Westermo Network Technologies AB, 2026). The Maroochy Shire sewage spill (2000) shows that insider threats with remote access can cause real physical damage. OT patching is uniquely difficult — systems run legacy software and cannot easily go offline (NIST, 2023). The 2024 Hezbollah pager incident further highlights how adversaries can compromise hardware long before it reaches an operational site.

What a Data Engineer Should Keep in Mind

Data Engineers in IT/OT environments must treat OT-sourced data with heightened care — corrupted sensor readings can directly affect safety-critical decisions. Secure ingestion pipelines, least-privilege access controls, anomaly detection, and continuous monitoring are all essential when connecting OT sources to analytics platforms (Westermo Network Technologies AB, 2026; NIST, 2023). Above all, a Data Engineer must understand the operational context behind the data — a pipeline that ignores OT constraints can inadvertently become an attack vector. Ultimately, IT/OT convergence demands a security mindset that combines technical rigour with operational awareness.

References

Westermo Network Technologies AB. (2026). *Holistic Security in Industrial Networks* [Guest lecture slides, Data Engineering II, Uppsala University, 5 May 2026].

NIST. (2023). *Guide to Operational Technology (OT) Security* (NIST Special Publication 800-82r3). <https://doi.org/10.6028/NIST.SP.800-82r3>